Solutions Discrete Probability Distributions

1. a. $f(2) = \binom{6}{2}(.23)^2(.77)^4 = .2789$

   b. $P(\text{at least } 2) = 1 - f(0) - f(1)$

   $$= 1 - \binom{6}{0}(.23)^0(.77)^6 - \binom{6}{1}(.23)^1(.77)^5$$

   $$= 1 - .2084 - .3735 = .4181$$

   c. $f(0) = \binom{10}{0}(.23)^0(.77)^{10} = .0733$

2. a. $f(x) = \binom{n}{x}(p)^x(1-p)^{n-x}$

   $$f(3) = \frac{10!}{3!(10-3)!}(.30)^3(1-.30)^{10-3}$$

   $$f(3) = \frac{10(9)(8)}{3(2)(1)}(.30)^3(1-.30)^7 = .2668$$

   b. $P(x \geq 3) = 1 - f(0) - f(1) - f(2)$

   $$f(0) = \frac{10!}{0!(10)!}(.30)^0(1-.30)^{10} = .0282$$

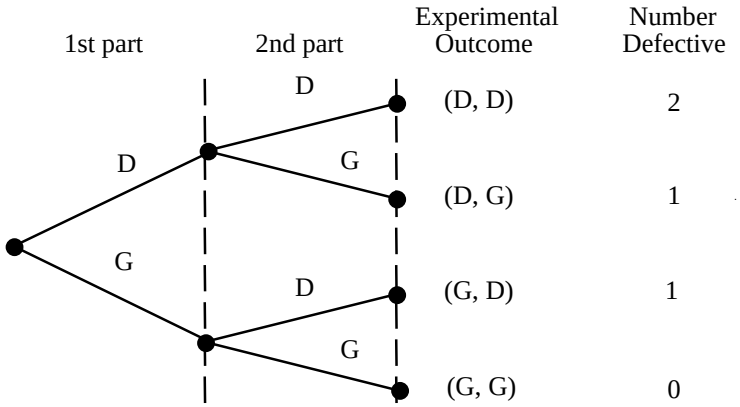
   $$f(1) = \frac{10!}{1!(9)!}(.30)^1(1-.30)^9 = .1211$$

   $$f(2) = \frac{10!}{2!(8)!}(.30)^2(1-.30)^8 = .2335$$

   $P(x \geq 3) = 1 - .0282 - .1211 - .2335 = .6172$

3. a. Probability of a defective part being produced must be .03 for each part selected; parts must be selected independently.

   b. Let: D = defective
   G = not defective



c. 2 outcomes result in exactly one defect.

d. P(no defects) = (.97) (.97) = .9409

P (1 defect) = 2 (.03) (.97) = .0582

P (2 defects) = (.03) (.03) = .0009

4. a. Yes. Since the employees are selected randomly, p is the same from trial to trial and the trials are independent. The two outcomes per trial are loyal and not loyal.

Binomial $n = 10$ and $p = .25$

$$f(x) = \frac{10!}{x!(10-x)!}(.25)^{x}(1-.25)^{10-x}$$

b. $$f(0) = \frac{10!}{0!(10-0)!}(.25)^{0}(1-.25)^{10-0} = .0563$$

c. $$f(4) = \frac{10!}{4!(10-4)!}(.25)^{4}(1-.25)^{10-4} = .1460$$

d. Probability $(x \geq 2) = 1 - f(0) - f(1)$

From part (b), $f(0) = .0563$

$$f(1) = \frac{10!}{1!(10-1)!}(.25)^{1}(1-.25)^{10-1} = .1877$$

Probability $(x \geq 2) = 1 - f(0) - f(1) = 1 - (.0563 + .1877) = .7560$

5. a. .90

b. P(at least 1) = $f(1) + f(2)$

$$f(1) = \frac{2!}{1!\,1!}(.9)^1(.1)^1$$

$$= 2(.9)(.1) = .18$$

$$f(2) = \frac{2!}{2!\,0!}(.9)^1(.1)^0$$

$$= 1(.81)(1) = .81$$

$\therefore P(\text{at least 1}) = .18 + .81 = .99$

Alternatively

$P(\text{at least 1}) = 1 - f(0)$

$$f(0) = \frac{2!}{0!\,2!}(.9)^0(.1)^2 = .01$$

Therefore, $P(\text{at least 1}) = 1 - .01 = .99$

c. $P(\text{at least 1}) = 1 - f(0)$

$$f(0) = \frac{3!}{0!\,3!}(.9)^0(.1)^3 = .001$$

Therefore, $P(\text{at least 1}) = 1 - .001 = .999$

d. Yes; $P(\text{at least 1})$ becomes very close to 1 with multiple systems and the inability to detect an attack would be catastrophic.

6. a. Using the 20 golfers in the Hazeltine PGA Championship, the probability that a PGA professional golfer uses a Titleist brand golf ball is $p = 12/20 = .6$

For the sample of 15 PGA Tour players, use a binomial distribution with $n = 15$ and $p = .6$.

$$f(10) = \frac{15!}{10!5!}(.6)^{10}(1-.6)^{15-10} = .1859$$

Or, using the binomial tables, $f(10) = .1859$

b. $P(x \geq 10) = f(11) + f(12) + f(13) + f(14) + f(15)$

Using the binomial tables, we have

$.1268 + .0634 + .0219 + .0047 + .0005 = .2173$

c. $E(x) = np = 15(.6) = 9$

7. a. $f(0) + f(1) + f(2) = .0115 + .0576 + .1369 = .2060$

b. $f(4) = .2182$

c. $1 - [f(0) + f(1) + f(2) + f(3)] = 1 - .2060 - .2054 = .5886$

d. $\mu = np = 20(.20) = 4$

8. a. $p = 1/4 = .25$

$$f(x) = \binom{n}{x}(p)^x(1-p)^{n-x}$$

$$f(4) = \binom{20}{4}(.25)^4(1-.25)^{20-4}$$

$$f(4) = \frac{20!}{4!(20-4)!}(.25)^4(.75)^{16} = f(4) = \frac{20(19)(18)(17)}{4(3)(2)(1)}(.25)^4(.75)^{16} = .1897$$

b. $P(x \geq 2) = 1 - f(0) - f(1)$

Using the binomial tables $f(0) = .0032$ and $f(1) = .0211$
$P(x \geq 2) = 1 - .0032 - .0211 = .9757$

c. Using the binomial tables $f(12) = .0008$

And, with $f(13) = .0002$, $f(14) = .0000$, and so on, the probability of finding that 12 or more investors have exchange-traded funds in their portfolio is so small that it is highly unlikely that $p = .25$. In such a case, we would doubt the accuracy of the results and conclude that p must be greater than .25.

d. $\mu = np = 20(.25) = 5$